

Exercise 8

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1. Repeat the illustrative accept-reject simulation multiple times (e.g., $s = 1000$) using $r = 100$ and summarize the results in the form of a histogram or frequency polygon. Next, do the same experiment but with $r = 1000$ and $r = 5000$. Compare to the exact area and discuss your results. Hint: You can simultaneously draw $s \times r$ random numbers and then collect them in s groups (see package `{dplyr}` functions `group_by()` and `summarize()`).

```
# Set seed for replicability (this defines a starting number from which the sequence of random numbers
set.seed(35739)

norm_pdf <- data.frame(x = seq(0.2,
                             1.2,
                             0.01)) %>%
  mutate(f = dnorm(x,
                  mean = 0,
                  sd = 0.5))

# setting parameters
r <- 100
s <- 1000 # Number of simulations
xmin <- 0.2
xmax <- 1.2
fmax <- max(norm_pdf$f) # Max height of PDF

sim_pdf <- data.frame(
  sim_group = rep(1:s, each = r), # Create 1000 groups of 100 draws
  x = runif(s * r, xmin, xmax),    # All x-values
  fsim = runif(s * r, 0, fmax)     # All y-values
)

sim_pdf <- sim_pdf %>%
  mutate(
    f = dnorm(x, mean = 0, sd = 0.5),
    status = ifelse(fsim <= f, "accept", "reject")
  )

# Calculate acceptance rate per simulation group
accept_rates <- sim_pdf %>%
  group_by(sim_group) %>%
  summarize(accept_rate = sum(status == "accept") / r)

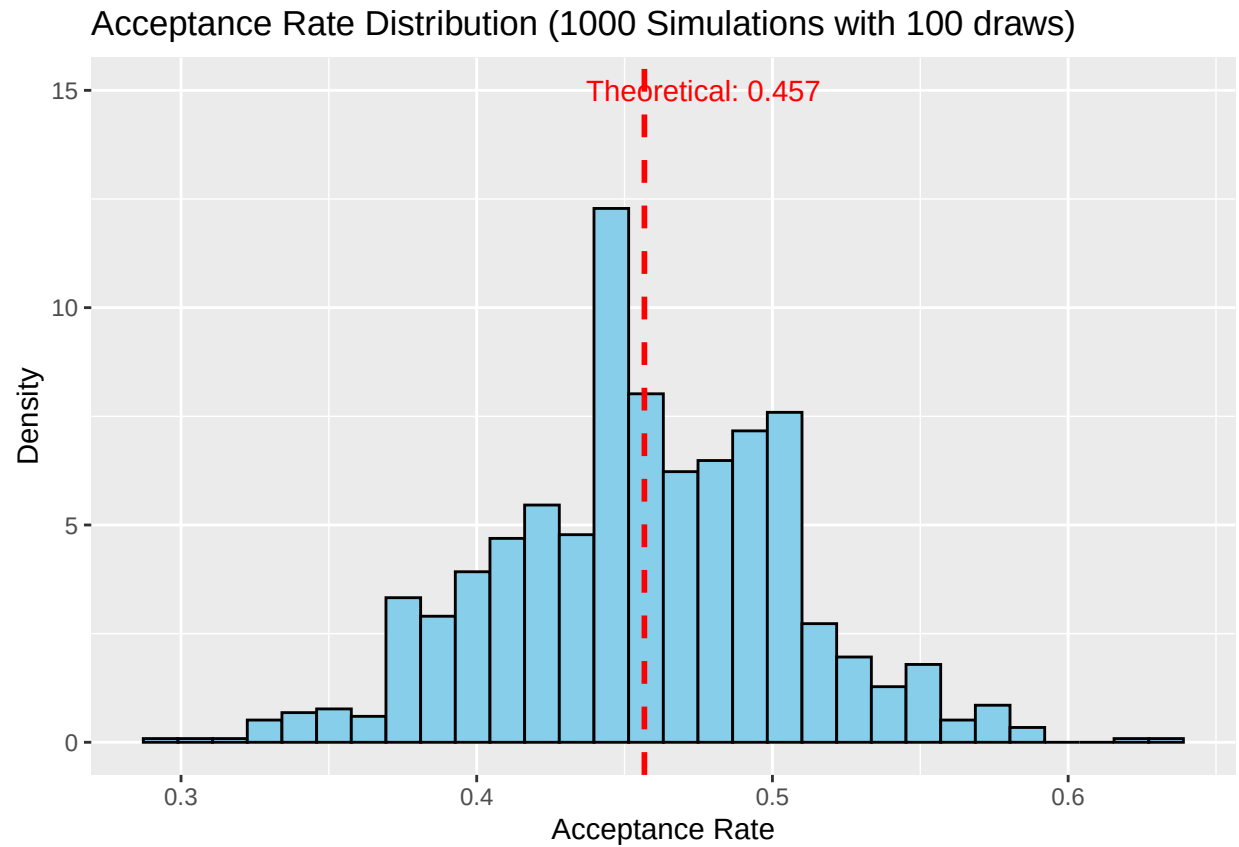
# Theoretical acceptance probability (for comparison)
theoretical_prob <- integrate(
```

```

function(x) dnorm(x, 0, 0.5),
lower = xmin,
upper = xmax
)$value / ((xmax - xmin) * fmax)

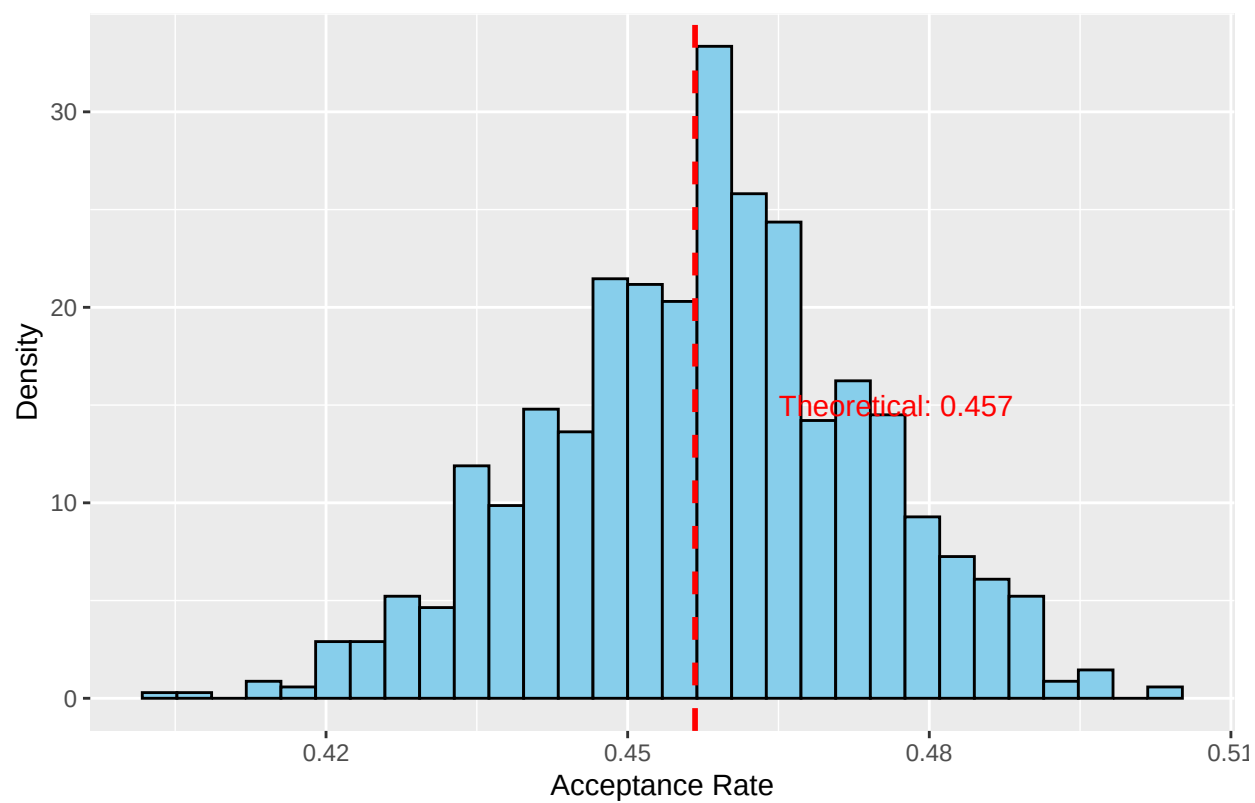
# Plot histogram of acceptance rates
ggplot(accept_rates, aes(x = accept_rate)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 30,
    fill = "skyblue",
    color = "black"
  ) +
  geom_vline(
    xintercept = theoretical_prob,
    color = "red",
    linetype = "dashed",
    linewidth = 1
  ) +
  labs(
    title = "Acceptance Rate Distribution (1000 Simulations with 100 draws)",
    x = "Acceptance Rate",
    y = "Density"
  ) +
  annotate(
    "text",
    x = theoretical_prob + 0.02,
    y = 15,
    label = paste0("Theoretical: ", round(theoretical_prob, 3)),
    color = "red"
  )

```

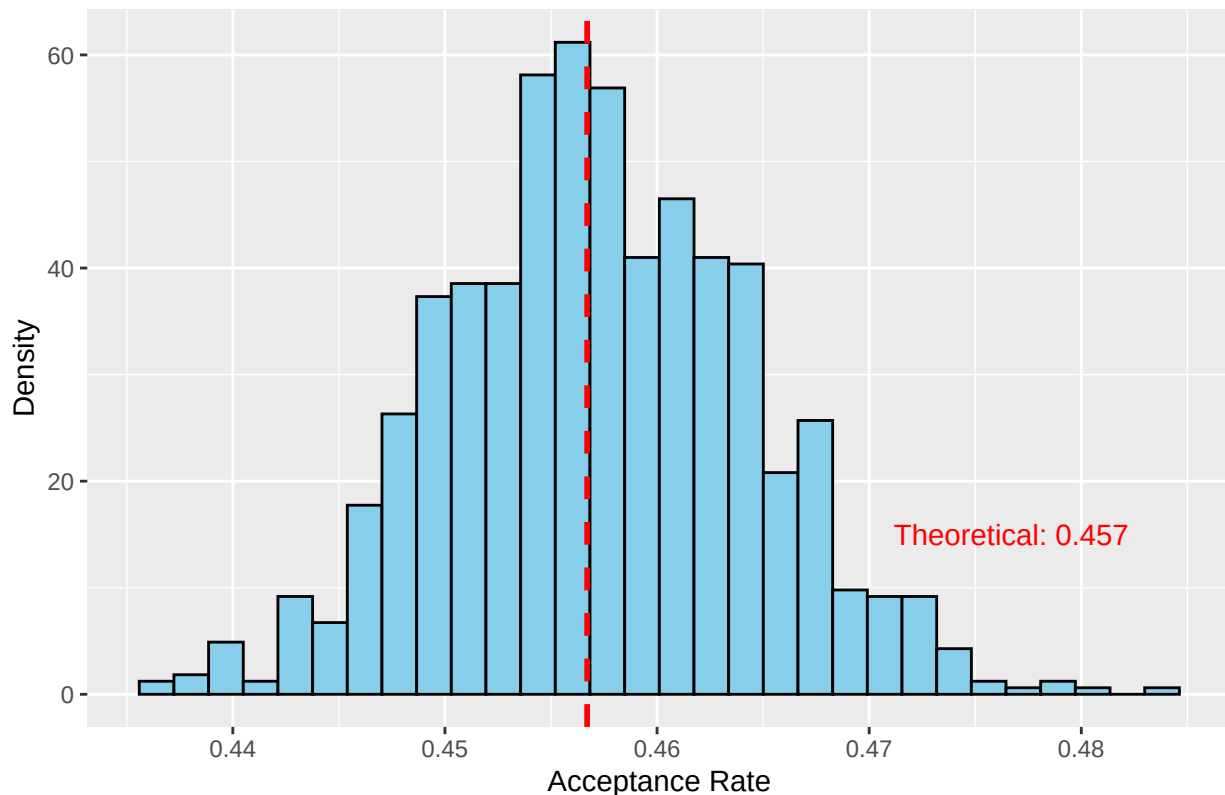


The codes for $r = 1000$ and $r = 5000$ are similar, the histograms for each are listed below.

Acceptance Rate Distribution (1000 Simulations with 1000 draws)



Acceptance Rate Distribution (1000 Simulations with 5000 draws)



The simulations demonstrate that increasing the number of draws per trial (r) significantly reduces variability in acceptance rate estimates. For $r = 100$, the histogram of acceptance rates across $s = 1000$ trials shows wider dispersion around the theoretical value than the other two trials. With $r = 1000$, the distribution narrows more, and for $r = 5000$, the histogram becomes more like normal distribution, being symmetric on the both sides theretical value line. This illustrates the **Law of Large Numbers** theory, which means that larger r yields more precise approximations of the true probability.

2. Re-estimate the multinomial probit model in this chapter, still with $r = 50$ but using a different random seed. Do this several times. The estimated values of the parameters will change, which is to be expected...but which ones change more? Why do you think this happens?

```
prob1 <- mlogit(depvar ~ ic + oc,
  H,
  reflevel = "gc",
  probit = TRUE,
  # can set the seed to control the random numbers (parameter1)
  seed = 3333,
  # number of draws (parameter2)
  R = 50)

prob2 <- mlogit(depvar ~ ic + oc,
  H,
  reflevel = "gc",
  probit = TRUE,
```

Table 1: Multinomial Probit models with different seeds

Variable	Seed = 3333		Seed = 1069		Seed = 2024	
	Estimate	p-value	Estimate	p-value	Estimate	p-value
(Intercept):ec	0.0075	0.9871	0.0471	0.9100	-0.0010	0.9984
(Intercept):er	0.0886	0.8690	0.1195	0.7662	0.1257	0.7932
(Intercept):gr	-48.6906	0.9315	-48.4593	0.6796	-21.5082	0.8507
(Intercept):hp	-1.5578	0.2578	-1.5869	0.1948	-1.7565	0.1823
ic	-0.0016	0.0175	-0.0017	0.0082	-0.0016	0.0109
oc	-0.0042	0.0024	-0.0042	0.0008	-0.0042	0.0011
ec.er	-0.0141	0.9904	0.0663	0.9411	0.0577	0.9558
ec.gr	9.2636	0.9378	9.3737	0.7207	3.2714	0.9000
ec.hp	-0.9114	0.4852	0.0694	0.9136	-0.6912	0.5274
er.er	1.1514	0.0389	1.2169	0.0125	1.2066	0.0673
er.gr	15.0825	0.9344	21.0297	0.6860	10.3419	0.8598
er.hp	-0.6594	0.6388	-0.4363	0.4331	-1.1299	0.2369
gr.gr	42.5679	0.9310	41.4606	0.6752	17.4803	0.8493
gr.hp	0.4390	0.7350	2.0776	0.1118	-0.4865	0.6398
hp.hp	0.7685	0.5719	0.0673	0.8727	0.4502	0.7157

Note:

Log-Lik (seed=3333):-1002.157; Log-Lik (seed=1069):-1001.638; Log-Lik (seed=2024):-1002.19

```

# can set the seed to control the random numbers (parameter1)
seed = 1069,
# number of draws (parameter2)
R = 50)

prob3 <- mlogit(depvar ~ ic + oc,
  H,
  reflevel = "gc",
  probit = TRUE,
  # can set the seed to control the random numbers (parameter1)
  seed = 2024,
  # number of draws (parameter2)
  R = 50)

```

Re-estimating the multinomial probit model with different seeds (`seed = 3333, 1069, 2024`) and `R = 50` reveals different patterns in parameter stability. First, The main coefficients (i.e., `ic` and `oc`) remain remarkably consistent across different runs. In contrast, intercepts and covariance parameters exhibit significant variability. For example, the intercept of `(Intercept):gr` ranges from -48.69 to -21.51, with uniformly insignificant p-values < 0.68 , indicating poor identification. Meanwhile, covariance terms like `ec.gr` (9.26 to 3.27) and `ec.er` (-0.01 to 0.07) vary widely, with p-values > 0.9 .

The reasons for can be concluded in several aspects: a) Limited draws. The Halton sequence approximation for the multivariate normal integrals becomes noisy with fewer draws, disproportionately affecting covariance terms and intercepts tied to latent utilities. b) Identification challenges. Covariance parameters and intercepts for rarely chosen alternatives (e.g., `gr`, `hp`) are poorly identified, amplifying sensitivity to simulation noise. One way to improve is to increase `R` for stable covariance estimation or simplify the model by restricting off-diagonal terms. This underscores the trade-off between computational cost and precision in multinomial probit models.

3. Use the models you estimated in Question 2 and repeat the scenario where the installation cost of gas central systems increases by 15%. What can you say about the patterns of substitution when you calculate them with different models? What would you do to be more confident about the estimated patterns of substitution?

Table 2: Predicted Choice Probabilities Under Different Policies (seed = 3333)

Policy	gc	ec	er	gr	hp
Do nothing	0.637	0.071	0.093	0.143	0.056
15% increase (seed = 3333)	0.574	0.095	0.118	0.144	0.071

Table 3: Predicted Choice Probabilities Under Different Policies (seed = 1069)

Policy	gc	ec	er	gr	hp
Do nothing	0.630	0.071	0.091	0.15	0.061
15% increase (seed = 1069)	0.567	0.094	0.117	0.15	0.074

Table 4: Predicted Choice Probabilities Under Different Policies (seed = 2024)

Policy	gc	ec	er	gr	hp
Do nothing	0.637	0.071	0.093	0.144	0.056
15% increase (seed = 2024)	0.574	0.094	0.117	0.145	0.070

Table 5: Substitution Effects between different models with different seeds

	gc	ec	er	gr	hp
Ratio of probabilities (seed = 3333)	-9.936	32.894	26.572	0.362	26.375
Ratio of probabilities (seed = 1069)	-9.838	33.325	27.488	0.208	23.448
Ratio of probabilities (seed = 2024)	-9.928	31.972	26.230	0.851	26.500

The substitution effects across models with different seeds (3333, 1069, 2024) reveal consistent trends but notable variability in magnitude. A 15% increase in gas central (**gc**) installation costs drives significant substitution toward electric central (**ec**) and electric room (**er**) systems, with their probability ratios increasing by ~30–33% and ~26–27%, respectively. However, substitutions to gas room (**gr**) and heat pump (**hp**) systems are smaller and more erratic, with **gr** fluctuating between 0.2–0.85%, and **hp** varying between 23–26%.

To improve confidence in the estimates, two ways could be implemented: a) Increase simulation draws. Raising **R** from 50 to more than 1,000 would reduce noise in the Halton sequence approximation, stabilizing covariance estimates and substitution patterns, though this would come with bigger computational costs. b) Robustness checks. Replicate the analysis across more seeds and models to distinguish true substitution trends from simulation artifacts.

4. Re-estimate the multinomial probit model in this chapter with $r = 50$ and the seed set to 3245, but change the reference level to “ec”. What happens? What do you think are the implications for experimenting with multinomial probit model specifications?

```
prob4 <- mlogit(depvar ~ ic + oc,
               H,
               refllevel = "ec",
               probit = TRUE,
               # can set the seed to control the random numbers (parameter1)
               seed = 3245,
               # number of draws (parameter2)
               R = 50)
```

Changing the reference level to “ec” (electric central) triggers a computational error regarding the “Cholesky decomposition of the covariance matrix”. The error **the leading minor of order 1 is not positive** indicates that the covariance matrix of the latent utilities is not positive definite when “ec” is the base category. This likely arises because the model’s covariance structure becomes near-singular due to collinearity between alternatives.

The implications are that when working with multinomial probit models, the choice of reference level can directly affect model results. Certain reference categories may introduce structural dependencies in the error covariance matrix, leading to estimation failures, therefore, experimentation with different reference levels is encouraged.